Akshay Vadher

Backend Engineer | Go, Node.js & Distributed Systems

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Professional Summary

Backend Engineer with 3+ years building high-throughput distributed systems using Go and Node.js. Experienced in designing scalable APIs, concurrent processing pipelines, asynchronous event-driven architectures, and performance-critical backend services for enterprise-scale platforms at ArcelorMittal Nippon Steel, Amdocs, and MSCI.

Technical Skills

Backend Tech: Node.js, Go, Python
Frameworks: Express.js, Gin, FastAPI
Backend & Concurrency: Goroutines, Channels, Worker Pools, Concurrent System Design, REST APIs, gRPC
Databases & Messaging: MySQL, MongoDB, Redis, Apache Kafka
Cloud, DevOps & OS: AWS (EC2, S3, RDS), Docker, Linux, CI/CD (Jenkins)
Core Skills: Microservices, Middleware and API Development, Scalable Backend Architecture, Distributed Systems, Caching Strategy

Professional Experience

ArcelorMittal Nippon Steel India (AM/NS India) — **Apr 2025 – Present**
Software Engineer — *Surat*

- *Owned design and delivery of a material-allocation microservice handling 50,000 orders/day across multi-dimensional constraints (grade, size, location); replaced a manual reconciliation process, cutting turnaround time 80% and eliminating 40+ hours of manual work weekly*
- *Designed a data pipeline ingesting demand forecasting, inventory, and production planning signals with async processing and conflict resolution logic; improved forecast accuracy by 15% across 6-week horizons.*
- *Collaborated with OMPartner (OMP) supply chain software team on project setup and data integration workflows, building connectors to synchronize demand, inventory, and production data between OMP and internal systems; ensured seamless integration for planning modules.*

Amdocs Development Centre India Pvt Ltd — **Sep 2022 – Jan 2025**
Software Developer — *Pune*

- *Architected billing processing modules in Go and C++ handling 1M+ accounts and 10M+ transactions daily, reducing processing latency by 25% for large-scale workflows*
- *Diagnosed and resolved critical-path bottlenecks in pipelines processing 10M+ transactions/day, reducing overhead by 20% and cutting billing error rates by 15%*
- *Built event-driven workflow automation with async job scheduling, cutting manual intervention by 30% and maintaining reliability during peak billing loads (5x normal volume)*

Morgan Stanley Capital International (MSCI) — **Jun 2021 – Dec 2021**
Software Developer Intern — *Pune*

- *Built data ingestion and processing pipelines for ESG & Climate analytics handling 1M+ data points daily — designed for throughput with batched processing and indexed storage for sub-second query response*
- *Engineered a data aggregation layer feeding real-time dashboards across 5,000 portfolios: optimized query execution and caching, reducing dashboard load time to sub-second.*

Projects

Rate-Limited API Gateway | Go, Gin, Middleware

- Designed and implemented an API gateway in Go/Gin supporting token-bucket rate limiting and concurrent request orchestration, handling 1000 req/sec in local load testing
- Built reusable middleware for authentication, structured logging, and request throttling — reduced integration time for new endpoints by reusing shared middleware chain

Distributed Notification & Alert System | Node.js, Express, Kafka, MongoDB, Redis — 🌐 GitHub

- Architected a multi-channel (Email/SMS/In-App) notification platform using Apache Kafka to decouple ingestion, processing, and delivery, enabling asynchronous, fault-tolerant handling with user preference management (channels, throttling limits, quiet hours, deduplication, aggregation).
- Optimized reliability and throughput by implementing worker-based consumers with retry logic, dead-letter queue support, Redis-backed rate limiting, and MongoDB logging — supporting thousands of notifications per second with reduced failure rates.

Periodic URL Health Monitor | Node.js, Express, WebSocket, MongoDB, Vanilla JS — 🔗 Live Demo | 🌐 GitHub

- Built a lightweight full-stack application that periodically pings a list of URLs, recording response times and availability, and streams live status updates to a browser dashboard using plain HTML, CSS, and JavaScript.
- Implemented backend services in Node.js/Express with scheduled health checks, MongoDB persistence for historical uptime data, and WebSocket APIs for real-time visualization; added retry logic and configurable thresholds to improve reliability.

Education

Sardar Vallabhbhai National Institute of Technology (SVNIT) — **2020 - 2022**
Master Of Technology (MTech) in Computer Engineering , GATE Qualified (Score: 527) — *Surat*
Gujarat Technological University — **2016 - 2020**
Bachelor of Engineering (B.E) in Computer Engineering — *Surat*